

Employment

Graduate Research Assistant **NC State University** **Apr 2025 – Present**

- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern **Isomer AI** **May 2025 – May 2026**

- Collaborated with product and design teams to ship 10+ production AI features: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows; 923 commits, 7 production deployments.
- Owned full-stack React/Node.js + TypeScript codebase: PostgreSQL, Redis, Docker, Auth0 RBAC, and Playwright test coverage.

Software Developer Intern **Mphasis (HP Company)** **Jun 2024 – Aug 2024**

- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern **Capgemini** **Jun 2023 – Aug 2023**

- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC **North Carolina State University** **Aug 2024 – May 2026**

- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India **Mumbai University** **Aug 2020 – May 2024**

- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

CortexQ **Kubernetes LLM Inference Platform** **2026**

- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability; 88 pytest tests.

VectorLift **Semantic Search & Ranking Engine** **2026**

- Built full retrieval pipeline in Python on MS MARCO (8.8M passages): BM25 baseline, dense bi-encoder fine-tuned with in-batch negatives, cross-encoder re-ranker; evaluated 6 pipeline configurations end-to-end.
- Scored NDCG@10, MRR@10, MAP with bootstrap tests; Kafka-driven Qdrant + Elasticsearch indexing; FastAPI serving with per-stage latency.

AlgoPulse **Full-Stack Agentic AI Platform** **2025**

- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load.
- Replaced polling with event-driven Supabase Edge Functions; sustained sub-200ms SSE delivery at 2,000 req/sec.

Languages and Technologies

- **Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- **AI & Search:** LangChain, RAG, Semantic Search, Elasticsearch, Qdrant, Sentence Transformers, LLM Evaluation, Claude API, GPT-4o
- **Infra & Backend:** Kubernetes, Docker, FastAPI, Node.js, Kafka, GCP, AWS, PostgreSQL, Redis, GitHub Actions
- **Full Stack & Observability:** React, Next.js, Auth0, OpenTelemetry, Prometheus, Grafana, Temporal.io, Playwright

Additional Experience and Awards

- **Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.
- **GDSC Lead:** Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.
- **Technical Writing:** Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.
- **IEEE Published:** Co-authored Computer Vision-Based Cybersecurity Threat Detection research; presented at Springer icSoftComp 2023 to 500+ global attendees.